Three positive point charges have at most four equilibria

A computer-assisted proof of Maxwell's bound for three charges

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Abstract

Maxwell conjectured in 1873 that the electrostatic field of n point charges in $\mathbb{R}^3$ has at most $(n-1)^2$ isolated equilibrium points. The conjecture is now known to fail for $n = 5$. For three positive charges the best bound in the literature is 6, obtained in 2026 by Gabrielov, Novikov, Novikov and Shapiro, who improved the earlier bound 12 of Gabrielov, Novikov and Shapiro. We prove that three positive Coulomb charges in $\mathbb{R}^3$ have at most four equilibria, which is Maxwell's bound for $n = 3$ and is sharp.

When the charges are not collinear, all equilibria lie in the open triangle they span, and none is a maximum of the potential restricted to the plane. When they are nondegenerate, each is a local minimum or a saddle, and there are $2 + 2m$ of them, where m is the number of minima. Using Lami's theorem, we write the charges that make a given point critical in closed form. A nondegenerate minimum must lie in the region $M = \{|\sum_i \lambda_i u_i^2| < \tfrac{1}{3}\}$, where λ_i are its barycentric coordinates and u_i the unit vectors from the charges (as complex numbers). The number of minima equals the winding number around q of a caustic curve C in the space of charge ratios. A Sturm-type lemma for plane fronts with three ordinary cusps, no inflections and total tangent turning π shows that this winding number is at most one in absolute value.

The hypotheses of the lemma reduce to four strict inequalities between explicit functions of four variables on a compact domain, plus one two-dimensional check for a single triangle. We verify them by branch and bound with outward-rounded interval arithmetic that does not rely on the platform's mathematical library. About 10^7 boxes were processed and none was left unresolved, in each of two different initial subdivisions. The computer-assisted part has not yet been independently checked; the accompanying source is provided for scrutiny.

Keywords: Maxwell's conjecture; point charges; equilibria of electrostatic fields; caustics; fronts; Sturm theory; interval arithmetic; computer-assisted proof.
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Status. This is a preprint and has not been peer reviewed. Sections 2–5 are mathematical arguments meant to be checked by hand. Theorem 5.1 is established by the computations of Section 7, whose correctness rests on the implementation described there. See Section 9 for what an independent check should cover, and the AI-assistance statement at the end.

1 Introduction

In his *Treatise on Electricity and Magnetism*, Maxwell [13] argued that the field of n point charges has at most $(n-1)^2$ points of equilibrium, provided they are isolated. His heuristic counted the intersections of the two curves on which two components of the field vanish. The statement became known as Maxwell's conjecture. It turned out to be remarkably hard even for $n = 3$.

Gabrielov, Novikov and Shapiro [5] brought the problem to modern attention. Using the theory of fewnomials [9], they proved general finiteness bounds. For three positive charges they showed that there are at most 12 nondegenerate equilibria. Zolotov [20] obtained further upper bounds via the Thom–Milnor theorem. Tsai [17] proved the bound 4 for three charges of equal magnitude, by a computer-assisted symbolic computation. Edelsbrunner, Fillmore and Oliveira [3] disproved a related conjecture of [5] that compares equilibria with Voronoi cells. In July 2026 two preprints appeared. Arathoon, Ball and Kvalheim [1] exhibited five charges whose field has at least $24 > 16 = (5-1)^2$ nondegenerate equilibria, so Maxwell's conjecture is false in general. Gabrielov, Novikov, Novikov and Shapiro [6] lowered the three-charge bound from 12 to 6. The question whether three positive charges can have 6 equilibria was therefore open.

We answer it for the Coulomb potential. The contribution is the geometric reduction for arbitrary positive charge ratios and the verified inequalities on the full shape domain. The support-function and Sturm-comparison tools belong to existing hedgehog theory, discussed in Section 4. Independently, Enciso and Peralta-Salas [4] proved finiteness for any number of charges of one sign; that result does not give the sharp three-charge bound used here.

Theorem 1.1. *Let $q_1, q_2, q_3 > 0$ be charges at three distinct points $p_1, p_2, p_3 \in \mathbb{R}^3$, and let $V(y) = \sum_i q_i |y - p_i|^{-1}$. Then the electrostatic field $E = -\nabla V$ vanishes at no more than four points. More precisely:*

(a) If the p_i are collinear, E has exactly two zeros.

(b) Otherwise every zero of E lies in the open triangle Δ° spanned by the charges. There is a closed curve $C = C(p_1, p_2, p_3)$ in the open simplex of normalized charge vectors such that, for $q \notin C$, the number of zeros is 2 or 4, according to whether q lies outside or inside C. Both cases occur for every triangle.

The proof is computer-assisted: it uses the interval-arithmetic verification of Theorem 5.1.

Since four equilibria occur (for instance for equal charges at the vertices of an equilateral triangle, a case already discussed by Maxwell), the bound is sharp. In the language of [6], Theorem 1.1 closes the gap between 6 and Maxwell's $(3-1)^2 = 4$ for positive Coulomb charges. In contrast with [5, 6], whose methods treat Riesz potentials of general exponent, we prove it only for the Coulomb exponent; see Section 8 for evidence about other exponents and about charges of mixed signs.

1.1 Strategy

The argument has a geometric part (Sections 2–5) and a computational part (Sections 6–7).

1. *Reduction.* For noncollinear sites and nondegenerate parameters, all equilibria lie in the open triangle Δ°. There they are critical points of the planar potential, never maxima, and their number is $2 + 2m$, where m is the number of local minima. So it suffices to show $m \le 1$.
2. *Inverse problem.* By Lami's theorem, a point $x \in \Delta^\circ$ is critical exactly for the charges $q \propto F(x) = (\lambda_i r_i^3)_i$. Here λ_i are the barycentric coordinates of x and $r_i = |x - p_i|$. The point is a nondegenerate minimum if and only if $|Z(x)| < \tfrac{1}{3}$, where $Z = \sum_i \lambda_i u_i^2$ in complex notation. This defines a purely geometric *minimum region* $M \subset \Delta^\circ$.
3. *Caustic.* The map F is the Gauss map of a surface, and M is its elliptic part. Hence the number of minima for the charges q equals $|\operatorname{wind}(C, q)|$, where $C = F(\partial M)$ is a caustic in the projective plane of charge ratios.

4. *A Sturm-type lemma.* A plane front with no inflections, total tangent turning π and exactly three ordinary cusps is a simple closed curve. Through every point pass at most three of its tangent lines, and its winding numbers lie in $\{0, \pm 1\}$ (Theorem 4.3).
5. *Verification.* The hypotheses of the lemma follow from four strict inequalities (J), (I), (T), (W) on a compact domain of four variables, plus a check on one triangle. The four inequalities are written in a local chart in which every quantity has a closed form (Section 6), and they are verified by interval arithmetic (Section 7).

1.2 On the computer-assisted part

The four inequalities involve rational functions of sin and cos of the angles of the configuration and of its barycentric coordinates. Every term has an explicit formula, derived in Section 6. We verify them on a compact four-dimensional domain using IEEE double precision. Every operation is rounded outward, and $\sin, \cos$ are enclosed by Taylor polynomials with explicit remainders, so no assumption is made on the accuracy of the platform's mathematical library. Enclosures are sharpened by mean-value forms computed with interval automatic differentiation. Two runs from different initial subdivisions each processed about 10^7 boxes and left none unresolved, in about eight minutes using 16 worker processes. The code is available [16]. Proofs that combine analysis with a rigorous computation over a compact parameter domain have a long history (see e.g. [7, 19]); for interval methods see [15, 18].

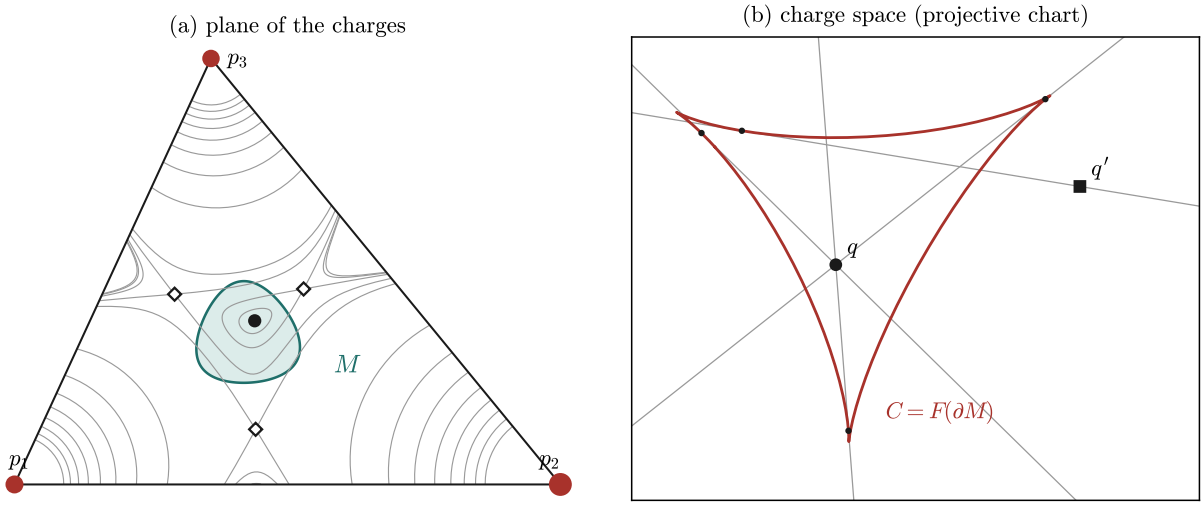


Figure 1. (a) Three charges (red, sized by magnitude) with q chosen inside the caustic. The equipotential lines are shown in grey. The minimum region $M = \{|Z| < \frac{1}{3}\}$ is shaded, and it contains the unique local minimum ($\bullet$). There are three saddles ($\diamond$), for four equilibria in total. (b) The caustic $C = F(\partial M)$ in a projective chart of charge space, a curvilinear triangle with three cusps. Exactly three tangent lines of C pass through the point q inside it, and one through the point q' outside. By Lemma 4.1 these counts give winding numbers 1 and 0, i.e. one minimum (four equilibria) for q and none (two equilibria) for q'.

2 Reduction to minima in the plane

Throughout, $s > 0$ is a Riesz exponent and $V(y) = \sum_i q_i |y - p_i|^{-s}$. The Coulomb case is $s = 1$, which corresponds to $s = 2\alpha$ in the notation of [5]. Nothing in Sections 2–4 depends on the value of s. Indices are taken modulo 3.

Let the charges be non-collinear, let Π be their plane, and let $\Delta \subset \Pi$ be the closed triangle with vertices p_1, p_2, p_3. We identify Π with $\mathbb{C}$ and write, for $x \in \Pi$,

$$r_i = |x - p_i|, \qquad u_i = \frac{x - p_i}{r_i}, \qquad \varphi_i(x) = r_i^{-s}, \qquad \lambda_i(x) = \text{barycentric coordinates},$$

so that $\sum_i \lambda_i = 1$ and $\sum_i \lambda_i (x - p_i) = 0$. We denote by $\theta_i \in (0, \pi)$ the angle at x subtended by the side $p_{i+1}p_{i+2}$, and by $\sigma_i = \sin\theta_i$; for $x \in \Delta^\circ$, $\theta_1 + \theta_2 + \theta_3 = 2\pi$. For complex numbers a, b we write $a \cdot b = \operatorname{Re}(a\bar{b})$ for the Euclidean inner product.

Lemma 2.1. *Let q_1, q_2, q_3 be nonzero, of arbitrary signs. In (a) and (b), assume that the sites p_i are non-collinear.*

(a) Every equilibrium lies in Π.

(b) If all $q_i > 0$, every equilibrium lies in Δ°.

(c) If all $q_i > 0$ and the p_i are collinear, there are exactly two equilibria.

Proof. (a) We have $\nabla V(y) = -s \sum_i w_i (y - p_i)$ with $w_i = q_i |y - p_i|^{-s-2} \neq 0$. Write $y = x + z\nu$, with $x \in \Pi$ and ν a unit normal to Π. If $\nabla V(y) = 0$ and $z \neq 0$, the normal component gives $\sum_i w_i = 0$, and then the tangential component gives $\sum_i w_i p_i = 0$. Since the p_i are affinely independent, $w = 0$, a contradiction. (b) If $x \in \Pi \setminus \Delta$, there is a unit vector e with $e \cdot (x - p_i) > 0$ for all i. Then $e \cdot \nabla V(x) = -s \sum_i w_i \, e \cdot (x - p_i) < 0$. If x lies in the relative interior of the side $p_{i+1}p_{i+2}$, take e normal to that side, pointing away from p_i; then $e \cdot (x - p_{i\pm1}) = 0$, $e \cdot (x - p_i) > 0$, and the same conclusion holds. (c) Off the line L of the charges, the component of ∇V orthogonal to L is a negative multiple of the offset from L, so it does not vanish. By rotational symmetry about L, a point of L is an equilibrium if and only if it is a critical point of $V|_L$. On L, as in (b), such points lie strictly between consecutive charges. On each of the two open segments, $V|_L$ is a sum of strictly convex functions and tends to $+\infty$ at both ends, so it has exactly one critical point. ∎

From now on, the charges are positive and non-collinear, and we study the critical points of the restriction $V_q := V|_\Pi$ in Δ°. The following parametrization is the classical statics of three concurrent forces.

Lemma 2.2 (Lami). *Let $x \in \Delta^\circ$ and $q \in \mathbb{R}^3_{>0}$. Then x is a critical point of V_q if and only if q is a positive multiple of*

$$F(x) := \big(\lambda_i(x)\, r_i(x)^{s+2}\big)_{i=1,2,3}. \tag{2.1}$$

Moreover $\lambda_i r_i = c\,\sigma_i$ with $c = r_1 r_2 r_3 / (2|\Delta|)$, so $F(x) \propto (\sigma_i r_i^{s+1})_i$.

Proof. We have $\nabla V_q = -s \sum_i w_i (x - p_i)$ with $w_i = q_i r_i^{-s-2}$. The weights $w \in \mathbb{R}^3_{>0}$ with $\sum_i w_i (x - p_i) = 0$ are exactly the positive multiples of $\lambda(x)$. For the second statement, note that $\lambda_i = \operatorname{area}(x, p_{i+1}, p_{i+2})/|\Delta| = \tfrac{1}{2} r_{i+1} r_{i+2} \sigma_i / |\Delta|$. ∎

Lemma 2.3. *At a critical point $x \in \Delta^\circ$ of V_q,*

$$\begin{aligned} \operatorname{Hess} V_q(x) &= s\, W \big[(s+2) T(x) - I\big], \\ W = \sum_i q_i r_i^{-s-2} > 0, &\qquad T(x) = \sum_i \lambda_i u_i u_i^{\mathsf{T}}. \end{aligned} \tag{2.2}$$

The eigenvalues of T are $\tfrac{1}{2}(1 \pm |Z|)$, where

$$Z(x) := \sum_{i=1}^{3} \lambda_i(x)\, u_i(x)^2 \in \mathbb{C}. \tag{2.3}$$

If $Z = |Z|e^{i\omega}$, the eigenvector of the smaller eigenvalue is $k = ie^{i\omega/2}$. Consequently, with $\kappa := s/(s+2)$ ($\kappa = \frac{1}{3}$ for Coulomb), the critical point is a nondegenerate local minimum if $|Z(x)| < \kappa$, a nondegenerate saddle if $|Z(x)| > \kappa$, and degenerate if $|Z(x)| = \kappa$. In the degenerate case, $\ker \operatorname{Hess} V_q(x) = \mathbb{R}k$ and the other eigenvalue is positive. It is never a local maximum.

Proof. $\nabla^2 r_i^{-s} = -s\, r_i^{-s-2}\big(I - (s+2)u_i u_i^{\mathsf T}\big)$, and $q_i r_i^{-s-2} = W\lambda_i$ by Lemma 2.2. Writing $u = e^{i\psi}$, we have $uu^{\mathsf T} = \frac{1}{2}I + \frac{1}{2}R(2\psi)$ with $R(\phi) = \left(\begin{smallmatrix}\cos\phi & \sin\phi \\ \sin\phi & -\cos\phi\end{smallmatrix}\right)$. Summing with weights λ_i gives $T = \frac{1}{2}I + \frac{1}{2}|Z|R(\omega)$. The eigenvalues of $\operatorname{Hess} V_q/(sW)$ are therefore $\frac{s+2}{2}(1 \pm |Z|) - 1$. The larger one is always positive, and the smaller one is positive, zero or negative according as $|Z| <$, $=$ or $> \kappa$. ∎

We call $M := \{x \in \Delta^\circ : |Z(x)| < \kappa\}$ the *minimum region*. It depends only on the triangle, and by the two lemmas above,

$$\begin{aligned} &\#\{\text{nondegenerate local minima of } V_q\} = \#\big(\hat F^{-1}(\hat q) \cap M\big), \\ &\hat F(x) := \frac{F(x)}{\sum_i F_i(x)}, \qquad \hat q := \frac{q}{\sum_i q_i}. \end{aligned} \tag{2.4}$$

Lemma 2.4. *If all critical points of V_q are nondegenerate, then $\#\text{saddles} = \#\text{minima} + 2$. Hence the number of equilibria is $2 + 2m$, where m is the number of local minima of V_q.*

Proof. Apply the Poincaré–Hopf theorem [14] to the gradient field ∇V_q on the sphere $\Pi \cup \{\infty\}$. Near each charge, ∇V_q is asymptotically radial, so it has index $+1$ there. Near ∞, $V_q \sim (\sum_i q_i)|x|^{-s}$ and the field is again radial, of index $+1$. Critical points in Π lie in Δ° and are minima (index $+1$) or saddles (index -1). Since $\chi(S^2) = 2 = 3 + 1 + \#\text{minima} - \#\text{saddles}$, the claim follows. ∎

Thus Theorem 1.1(b) is equivalent to the statement that V_q never has more than one local minimum, together with a separate treatment of degenerate configurations (Section 5).

3 The minimum region and the caustic

3.1 The map Z

Since $|u_i^2| = 1$, we have $|Z| \le 1$, with equality only if the u_i^2 with $\lambda_i > 0$ coincide; this never happens in Δ°. As complex functions of $z = x$, the Wirtinger derivatives of Z are written $Z_z, Z_{\bar z}$, and its Jacobian determinant as a map $\mathbb{R}^2 \to \mathbb{R}^2$ is

$$J(x) := |Z_z|^2 - |Z_{\bar z}|^2. \tag{3.1}$$

Lemma 3.1. *The map $Z : \Delta^\circ \to \mathbb{D} = \{|\zeta| < 1\}$ is proper and has degree -1 (with the standard orientations).*

Proof. Blow up each vertex p_i of Δ into the arc of directions $\psi \in [\arg(p_{i+1} - p_i), \arg(p_{i+2} - p_i)]$ (of length equal to the interior angle α_i). This gives a closed disk $\widetilde\Delta$ with interior Δ°. We claim that Z extends continuously to $\widetilde\Delta$ with boundary values of modulus one. On the relative interior of a side of direction $e^{i\beta}$, the two vectors u_j with $\lambda_j > 0$ are $\pm e^{i\beta}$, so $Z = e^{2i\beta}$. Approaching p_i in the direction ψ, we have $\lambda_i \to 1$ and $u_i \to e^{i\psi}$, so $Z \to e^{2i\psi}$. These values match at the endpoints of

each arc, and the convergence is uniform. Hence $|Z(x)| \to 1$ as $x \to \partial\Delta$, so Z is proper and its degree is the winding number of $Z|_{\widetilde{\partial\Delta}}$ around 0. Traverse $\widetilde{\partial\Delta}$ counterclockwise. Z is constant along the sides. At the vertex p_i the direction ψ runs through the interior angle clockwise, so $\arg Z$ decreases by $2\alpha_i$. The total change of $\arg Z$ is $-2(\alpha_1+\alpha_2+\alpha_3) = -2\pi$. ∎

The first condition to be verified by computer is

$$\textbf{(J)} \quad J(x) < 0 \text{ for every } x \in \Delta^\circ \text{ with } |Z(x)| \le \kappa.$$

Proposition 3.2. *Assume (J). Then Z maps $\overline{M}$ diffeomorphically and orientation-reversingly onto the closed disk $\{|\zeta| \le \kappa\}$. In particular, $\overline{M} \subset \Delta^\circ$ is a closed disk, and ∂M is a smooth Jordan curve, parametrized by*

$$x_0(\omega) := Z^{-1}\big(\kappa e^{i\omega}\big) \in \partial M, \qquad \omega \in \mathbb{R}/2\pi\mathbb{Z}. \tag{3.2}$$

As ω increases, ∂M is traversed clockwise, and $t(\omega) := dx_0/d\omega \neq 0$.

Proof. By (J), every preimage of a point ζ with $|\zeta| \le \kappa$ has local degree -1. By Lemma 3.1 the number of such preimages is $|\deg Z| = 1$. So Z restricts to a bijective local diffeomorphism $Z^{-1}(\{|\zeta| \le \kappa\}) \to \{|\zeta| \le \kappa\}$, which is a diffeomorphism. Since Z is a local diffeomorphism there, $Z^{-1}(\{|\zeta| \le \kappa\}) = \overline{M}$. Orientation reversal turns the counterclockwise circle into a clockwise curve. ∎

Differentiating $Z(x_0(\omega)) = \kappa e^{i\omega}$ gives $Z_z t + Z_{\bar z}\bar t = iZ$. Solving this real-linear equation,

$$t = \frac{\overline{Z_z}\,(iZ) - Z_{\bar z}\,\overline{(iZ)}}{J}. \tag{3.3}$$

3.2 The charge map and the caustic

Let $A = \{q \in \mathbb{R}^3 : \sum_i q_i = 1\}$ and let $\Sigma = A \cap \mathbb{R}^3_{>0}$ be the open simplex of normalized charges. By Lemma 2.2, $\hat F : \Delta^\circ \to \Sigma$ sends each point to the unique normalized charge vector for which it is critical. Geometrically, put $\Phi = (\varphi_1, \varphi_2, \varphi_3) : \Delta^\circ \to \mathbb{R}^3$. Then $V_q = \langle q, \Phi\rangle$ is a height function on the surface $\Phi(\Delta^\circ)$, and $\hat F$ is its Gauss map (see [2] for Gauss maps, caustics and fronts). By Lemma 2.3, M is the elliptic part of the surface and ∂M its parabolic curve. We call

$$C := \hat F(\partial M) = \{c(\omega) := \hat F(x_0(\omega)) : \omega \in \mathbb{R}/2\pi\mathbb{Z}\} \subset \Sigma \tag{3.4}$$

the *caustic*. It is the set of normalized charges for which V_q has a degenerate critical point.

Lemma 3.3. *$\hat F$ is a local diffeomorphism at every $x \in \Delta^\circ \setminus \partial M$. The sign of $\det D\hat F$ is constant on M, and $D\hat F(x_0)$ has rank one with kernel $\mathbb{R}k$ at every $x_0 \in \partial M$.*

Proof. Let $G(x,q) = \sum_i q_i \nabla\varphi_i(x)$, and let $B_x : \mathbb{R}^3 \to \mathbb{R}^2$, $B_x(\dot q) = \sum_i \dot q_i \nabla\varphi_i(x)$. Since any two of the vectors $\nabla\varphi_i(x) \parallel u_i$ are independent, B_x has rank 2, and its kernel is spanned by $F(x) \notin TA = \{\sum \dot q_i = 0\}$. Hence $B_x|_{TA}$ is an isomorphism, and its determinant has constant sign on the connected set Δ°. Differentiating $G(x, \hat F(x)) = 0$ gives $H_x + B_x \circ D\hat F(x) = 0$ with $H_x = \operatorname{Hess} V_{\hat F(x)}(x)$. So $\det D\hat F = \det H_x / \det(B_x|_{TA})$, and $\ker D\hat F = \ker H_x$. Lemma 2.3 finishes the proof. ∎

Proposition 3.4. *Assume (J). For every $\hat q \in \Sigma \setminus C$,*

$$\#\{\text{local minima of } V_q\} = \big|\operatorname{wind}(C, \hat q)\big|. \tag{3.5}$$

Proof. By Proposition 3.2, $\overline{M}$ is a closed disk. By Lemma 3.3, every preimage of $\hat q$ in M contributes the same sign ϵ to the degree of $\hat F|_{\overline{M}}$. Hence $\epsilon \cdot \#(\hat F^{-1}(\hat q) \cap M) = \deg(\hat F, M, \hat q) = \operatorname{wind}(\hat F|_{\partial M}, \hat q)$, and we conclude by (2.4). ∎

3.3 Tangent lines, cusps and inflections of the caustic

Let $k(\omega) = ie^{i\omega/2}$ be the kernel direction at $x_0(\omega)$ (Lemma 2.3). Define the *dual vector* $m(\omega) \in \mathbb{R}^3$ by

$$m_i(\omega) := \partial_k \varphi_i\big(x_0(\omega)\big) = -s\, r_i^{-s-1}\,(u_i \cdot k). \tag{3.6}$$

Lemma 3.5. *For every ω, $m(\omega) \neq 0$, and the line*

$$L(\omega) := \{q \in A : \langle q, m(\omega)\rangle = 0\} \tag{3.7}$$

contains $c(\omega)$ and $c'(\omega)$ is parallel to it. Thus C is a front with tangent lines $L(\omega)$. Moreover $m(\omega + 2\pi) = -m(\omega)$. Finally, $c'(\omega) = 0$ if and only if $t(\omega) \parallel k(\omega)$; these are the cusps of C.

Proof. All three $u_i \cdot k$ cannot vanish, since the u_i are pairwise non-parallel; so $m \neq 0$. Let $q(\omega) = F(x_0(\omega))$. Then $\langle q, m\rangle = \partial_k V_q(x_0) = 0$ because x_0 is critical. Differentiating $G(x_0(\omega), q(\omega)) = 0$ gives $\sum_i q_i' \nabla\varphi_i + H\,t = 0$. Taking the inner product with $k \in \ker H$ yields $\langle q', m\rangle = 0$. In the affine chart A this says that $c \in L$ and $c' \parallel L$. The identity $m(\omega + 2\pi) = -m(\omega)$ holds because $k(\omega + 2\pi) = -k(\omega)$. Finally, $c' = D\hat{F}(x_0)\,t$, which vanishes if and only if $t \in \ker D\hat{F} = \mathbb{R}k$ (Lemma 3.3). ∎

Because of the sign flip of m, the dual curve $\omega \mapsto [m(\omega)]$ is a one-sided closed curve in the projective plane. In particular, the normal direction of $L(\omega)$ turns by an odd multiple of π along one circuit of C. A *tangent line of C through $\hat{q}$* is a line $L(\omega)$ containing $\hat{q}$, i.e. a zero of $\omega \mapsto \langle q, m(\omega)\rangle$.

We now fix Euclidean coordinates on A. Write $m = \mathring{m} + \mu\mathbf{1}$ with $\mathring{m} \perp \mathbf{1}$, where $\mathbf{1} = (1,1,1)$. The direction of $L(\omega)$ is then determined by $\mathring{m}(\omega)$. We have $\mathring{m} \neq 0$, because $\mathring{m} = 0$ would make $L(\omega)$ the line at infinity $\{\sum q_i = 0\}$, which misses $\Sigma \ni c(\omega)$. Let $\vartheta(\omega)$ be a continuous determination of the angle of $\mathring{m}(\omega)$. We define the *total turning* of C as

$$\Theta := \vartheta(\omega + 2\pi) - \vartheta(\omega) \in \pi(2\mathbb{Z} + 1). \tag{3.8}$$

An *inflection* of C is a parameter where the tangent line is stationary, $m' \parallel m$.

Lemma 3.6. *$m \times m' = \gamma(\omega)\, q(\omega)$ for a continuous scalar γ, and $\vartheta'(\omega) \neq 0$ if and only if $\gamma(\omega) \neq 0$, i.e. if and only if ω is not an inflection.*

Proof. From $\langle q, m\rangle = 0$ and $\langle q', m\rangle = 0$ we get $\langle q, m'\rangle = 0$. So m and m' are both orthogonal to q, and $m \times m' \parallel q$. In an orthonormal basis $e_1, e_2, \mathbf{1}/\sqrt{3}$ of $\mathbb{R}^3$ with $e_1, e_2 \perp \mathbf{1}$, write $\mathring{m} = \alpha e_1 + \beta e_2$. Then $\det(m, m', \mathbf{1}/\sqrt{3}) = \alpha\beta' - \beta\alpha' = |\mathring{m}|^2 \vartheta'$. Hence $|\mathring{m}|^2\vartheta' = \gamma\langle q, \mathbf{1}\rangle/\sqrt{3}$, and $\langle q, \mathbf{1}\rangle > 0$. ∎

4 A Sturm-type lemma for fronts

The support-function description of fronts is classical; see [10, 11]. The three-tangent bound of Lemma 4.2(a) is essentially known. Martinez-Maure's Sturm-type comparison theorem [12] states that a 2π-periodic function h has at most as many zeros in $[0, 2\pi)$ as $h + h''$, plus two. For q off the curve, apply it to g_q below: $g_q + g_q'' = -\rho$ has six zeros in $[0, 2\pi)$, so g_q has at most eight, hence at most four in $[0, \pi)$ by antiperiodicity, and at most three by the parity in Lemma 4.1. We give a self-contained elementary proof. It also covers points q on the curve and shows that g_q cannot have two distinct double zeros (Lemma 4.2(b)). That refinement is what gives the simplicity of C.

This section is pure plane geometry. Identify $\mathbb{R}^2$ with $\mathbb{C}$, let $n(\theta) = e^{i\theta}$, $n'(\theta) = ie^{i\theta}$, and fix $\Theta \in \pi(2\mathbb{Z}+1)$ with $\Theta > 0$. Let $h \in C^\infty(\mathbb{R})$ satisfy $h(\theta+\Theta) = -h(\theta)$. The associated curve and its *radius-of-curvature function* are

$$x(\theta) = h(\theta)\,n(\theta) + h'(\theta)\,n'(\theta), \qquad \rho = h + h''. \tag{4.1}$$

Then $x(\theta+\Theta) = x(\theta)$ and $x' = \rho\, n'$. The line $\{y : y\cdot n(\theta) = h(\theta)\}$ contains $x(\theta)$ and is its tangent line. Cusps of x correspond to zeros of ρ. Such curves are the (multi)hedgehogs of Langevin, Levitt and Rosenberg [10], and h is their support function. For $q \in \mathbb{C}$ put

$$\begin{aligned} g_q(\theta) = q\cdot n(\theta) - h(\theta), \qquad g_q + g_q'' = -\rho, \qquad g_q(\theta+\Theta) = -g_q(\theta), \\ x(\theta) - q = -\bigl(g_q + i g_q'\bigr)\, n(\theta). \end{aligned} \tag{4.2}$$

The zeros of g_q are the parameters of the tangent lines through q, and $x(\theta) = q$ if and only if $g_q(\theta) = g_q'(\theta) = 0$.

Lemma 4.1 (counting tangent lines). *Let $q \notin x(\mathbb{R})$. Then the zeros of g_q are simple, their number N_q in $[0,\Theta)$ is odd, and*

$$\operatorname{wind}\bigl(x|_{[0,\Theta]}, q\bigr) = \frac{1}{2}\Bigl(\frac{\Theta}{\pi} - N_q\Bigr). \tag{4.3}$$

Proof. Put $\Gamma = g_q + i g_q'$, which never vanishes. By (4.2), $\arg(x-q) = \theta + \pi + \arg\Gamma$, so the change of $\arg(x-q)$ over $[0,\Theta]$ is $\Theta + \Delta\arg\Gamma$. Since $\Gamma(\theta+\Theta) = -\Gamma(\theta)$, the function g_q changes sign an odd number of times per period. Its zeros are simple, because a double zero would put q on the curve. Let $a < b$ be consecutive zeros. On (a,b), Γ stays in one open half-plane $\{\operatorname{Re}\Gamma \gtrless 0\}$. If $\operatorname{Re}\Gamma > 0$ there, then $g_q'(a) > 0 > g_q'(b)$, so Γ runs from the positive to the negative imaginary axis through the right half-plane. If $\operatorname{Re}\Gamma < 0$, it runs from the negative to the positive imaginary axis through the left half-plane. In both cases $\Delta\arg\Gamma = -\pi$ on $[a,b]$. The pieces $[0,a_1]$ and $[a_N,\Theta]$ together contribute $-\pi$ as well, by antiperiodicity. Hence $\Delta\arg\Gamma = -N_q\pi$, and the winding number is $(\Theta - N_q\pi)/2\pi$. $\blacksquare$

Lemma 4.2 (Sturm bound). *Let $\Theta = \pi$, and assume that ρ has exactly three zeros in $[0,\pi)$, all simple. Then for every $q \in \mathbb{C}$:*

(a) g_q has at most three zeros in $[0,\pi)$;

(b) g_q does not have two distinct double zeros ($g_q = g_q' = 0$) in $[0,\pi)$.

Proof. Let c be a zero of ρ, and let $a < b$ be the other two zeros in $(c, c+\pi)$. Write $g = g_q$. On $(c, c+\pi)$ put $f = g/\sin(\theta - c)$ and

$$\begin{aligned} \Psi = g'\sin(\theta-c) - g\cos(\theta-c), \qquad f' = \frac{\Psi}{\sin^2(\theta-c)}, \\ \Psi' = (g + g'')\sin(\theta-c) = -\rho\,\sin(\theta-c). \end{aligned}$$

So Ψ is strictly monotone on $[c,a]$, $[a,b]$ and $[b,c+\pi]$, with alternating senses. Also $\Psi(c) = -g(c)$, and $\Psi(c+\pi) = g(c+\pi) = -g(c)$. Suppose the senses are increasing, decreasing, increasing (the other case is symmetric), and let $w_0 = -g(c)$. Then $\Psi > w_0$ on $(c,a]$ and $\Psi < w_0$ on $[b,c+\pi)$. If $w_0 > 0$, Ψ has no zero in $(c,a]$ and at most one on each of the other two pieces. If $w_0 < 0$, it has at most one on each of the first two pieces and none on the third. If $w_0 = 0$, it has exactly one, on (a,b). Hence

$$\Psi \text{ has at most two zeros in } (c,c+\pi), \text{ and at most one if } g(c) = 0.$$

(a) Between consecutive zeros of f there is a zero of f', i.e. of Ψ. If $g(c) \neq 0$, then g has at most $2 + 1 = 3$ zeros in $(c, c+\pi)$ and none at c. If $g(c) = 0$, it has at most $1 + 1 = 2$ zeros in $(c, c+\pi)$, plus c. (b) Suppose $\theta_1 \neq \theta_2$ are double zeros in $[c, c+\pi)$. If neither equals c, then f has double zeros at $\theta_1 < \theta_2$. So f' vanishes at θ_1, at θ_2 and in between, giving three zeros of Ψ, which contradicts the bound 2. If $\theta_1 = c$, then $g(c) = g'(c) = 0$, so $f(\theta) \to 0$ as $\theta \to c^+$ and as $\theta \to (c+\pi)^-$. With $f(\theta_2) = f'(\theta_2) = 0$, Rolle's theorem on $[c, \theta_2]$ and $[\theta_2, c+\pi]$ gives three zeros of Ψ, which contradicts the bound 1. ∎

Theorem 4.3 (three cusps, three tangents). *Let $\Theta = \pi$, and assume that ρ has exactly three zeros per period, all simple. Then:*

(i) at most three tangent lines of x pass through any point of the plane;

(ii) $x : \mathbb{R}/\pi\mathbb{Z} \to \mathbb{C}$ is injective, i.e. the curve is a simple closed curve with three ordinary cusps;

(iii) $\operatorname{wind}(x, q) \in \{0, -1\}$ for every $q \notin x(\mathbb{R})$. It equals -1 if three tangent lines pass through q, and 0 if only one does.

Proof. (i) is Lemma 4.2(a). (iii) follows from Lemma 4.1, since N_q is odd and at most 3. For (ii), a double point $P = x(\theta_1) = x(\theta_2)$ with $\theta_1 \not\equiv \theta_2 \pmod{\pi}$ would give two double zeros of g_P, which Lemma 4.2(b) excludes. ∎

The orientation convention here (increasing θ) may differ from that of ω on the caustic. Since only $|\operatorname{wind}|$ enters, this does not matter.

5 The verified conditions and the proof of the main theorem

Let Δ be a positively oriented triangle; the negatively oriented case follows by reflection. Besides (J), we use the following conditions at points of ∂M. Here t is given by (3.3), and $\nu = d \arg t / d\omega$ is the turning rate of the tangent of ∂M.

- **(I)** $\det\big(\mathbf{1}, m(\omega), m'(\omega)\big) \neq 0$ for all ω. By Lemma 3.6 and the identity $\det(\mathbf{1}, m, m') = \langle \mathbf{1}, m \times m' \rangle = \gamma \langle \mathbf{1}, q \rangle$, this says that C has no inflections.
- **(T)** $\nu(\omega) < \frac{1}{2}$ at every ω with $t(\omega) \parallel k(\omega)$.
- **(W)** For the vertex labelled p_1 in the frame of Section 6 (in which θ_3 is the largest of the three angles at x_0), $\big| d \arg u_1(x_0(\omega))/d\omega \big| < \frac{1}{2}$.

Theorem 5.1 (computer-assisted). *Let $s = 1$. For every positively oriented triangle Δ:*

1. *(J) holds on $\overline{M}$, and (I) and (T) hold on ∂M;*
2. *(W) holds at every point $x_0 \in \partial M$, for the vertex that the chart of Section 6 labels p_1 at x_0;*
3. *for the base triangle Δ_0 with vertices 0, 1, $\frac{1}{2} + \frac{i}{8}$, the angle subtended by the side $[0,1]$ is the strictly largest of the three angles at every point of $\overline{M}(\Delta_0)$. Consequently the labelling in 2 is the same at all points of $\partial M(\Delta_0)$, and $|d \arg u_1 / d\omega| < \frac{1}{2}$ holds along all of $\partial M(\Delta_0)$ for the fixed vertex $p_1 = 0$.*

Items 1–2 are verified in Section 7 on a compact four-dimensional domain that covers all pairs (Δ, x_0), and item 3 by a separate two-dimensional computation. We now derive Theorem 1.1 from Theorem 5.1. Assume (J) throughout, so that Proposition 3.2 applies.

Proposition 5.2. *Under (J), (I) and (T), the caustic C has exactly three cusps per circuit. In the parametrization by the tangent direction ϑ, its radius-of-curvature function has exactly three simple zeros per period.*

Proof. Let $\beta(\omega) = \arg t - \arg k$ (continuous determination). By Proposition 3.2, ∂M is a smooth Jordan curve traversed clockwise, so $\arg t$ decreases by 2π over one circuit (Hopf's Umlaufsatz). Since $\arg k = \frac{\pi}{2} + \frac{\omega}{2}$ increases by π, β decreases by 3π. By Lemma 3.5, cusps are the parameters where $\beta \in \pi\mathbb{Z}$. At these, (T) gives $\beta' = \nu - \frac{1}{2} < 0$. A continuous function that can cross each level $j\pi$ only downward, and that decreases by 3π in total, meets the set $\pi\mathbb{Z}$ exactly three times per period.

For simplicity, write $t = |t|(\cos\beta\, k + \sin\beta\, ik)$. Since $D\hat{F}(x_0)k = 0$, we get $c'(\omega) = |t| \sin\beta\; D\hat{F}(x_0)(ik)$, where $D\hat{F}(x_0)(ik) \neq 0$ is parallel to $L(\omega)$ by Lemmas 3.3 and 3.5. By (I) and Lemma 3.6, ϑ is a strictly monotone function of ω. Reparametrize C by ϑ, so that C takes the form (4.1) with $dc/d\omega = \rho(\vartheta)\,\vartheta'(\omega)\,n'(\vartheta)$. Then $\rho(\vartheta(\omega))\,\vartheta'(\omega) = \pm|t| \sin\beta\, |D\hat{F}(x_0)(ik)|$. The right-hand side has simple zeros exactly where $\beta \in \pi\mathbb{Z}$, because $\beta' \neq 0$ there, and $\vartheta' \neq 0$. ∎

Here we used that, once $\vartheta' \neq 0$, C admits the representation (4.1). Explicitly, let $h(\vartheta)$ be the signed distance of L from the origin of the Euclidean chart. Then $c \cdot n = h$, and differentiating together with $c' \parallel L$ gives $c \cdot n' = h'$. The antiperiodicity $h(\vartheta + \Theta) = -h(\vartheta)$ expresses the fact that the line returns with opposite normal.

Proposition 5.3. *Under (J) and (I), $|\Theta|$ is the same for all positively oriented triangles. Under item 3 of Theorem 5.1, $|\Theta(\Delta_0)| = \pi$. Hence $|\Theta| = \pi$ for every triangle.*

Proof. Positively oriented triangles modulo similarities form a connected space, for instance $p_1 = 0$, $p_2 = 1$, $\operatorname{Im} p_3 > 0$. By (J) and the implicit function theorem, $x_0(\omega)$ depends smoothly on (ω, Δ), and hence so do m and $\mathring{m} \neq 0$. So $\Theta(\Delta) \in \pi(2\mathbb{Z}+1)$ depends continuously on Δ and is constant.

For Δ_0, consider the vertex $e_1 = (1,0,0)$ of the closed simplex. Since Σ is convex and does not contain e_1, we have $\operatorname{wind}(C, e_1) = 0$. The tangent lines through e_1 are the zeros of $\langle e_1, m\rangle = m_1 \propto u_1 \cdot k = \cos(\arg u_1 - \arg k)$. Along $\partial M(\Delta_0)$, $\arg u_1$ has net change 0, because p_1 lies outside the Jordan curve ∂M. Since $\arg k$ increases by π, the difference $\arg u_1 - \arg k$ decreases by exactly π, strictly monotonically by item 3. So it meets $\frac{\pi}{2} + \pi\mathbb{Z}$ exactly once, and $N_{e_1} = 1$. Lemma 4.1 (with C oriented so that ϑ increases) gives $0 = \frac{1}{2}(|\Theta|/\pi - 1)$. ∎

Proof of Theorem 1.1. Part (a) and the location statement are Lemma 2.1. Let Δ be non-degenerate; by reflection we may assume it positively oriented. By Theorem 5.1 and Propositions 5.2 and 5.3, the caustic C, parametrized by its tangent direction, satisfies the hypotheses of Theorem 4.3. Hence C is a simple closed curve and $|\operatorname{wind}(C, \hat{q})| \leq 1$ for $\hat{q} \notin C$.

Generic charges. If $\hat{q} \notin C$, all critical points of V_q are nondegenerate. By Proposition 3.4 there is at most one minimum, and by Lemma 2.4 there are 2 or 4 equilibria. The value is 4 exactly when $\hat{q}$ lies inside C. Both cases occur: C is a compact subset of Σ, so points of Σ near its corners lie outside C. And $M \neq \emptyset$, because $\deg Z \neq 0$ gives a point with $Z = 0$, so $\hat{F}(M)$ is a nonempty open subset of the interior of C.

At most one critical point in $\overline{M}$. If $\hat{q} \notin C$, this is the previous paragraph. Let $\hat{q} \in C$, and suppose that $\hat{F}(x) = \hat{q}$ for some $x \in M$. Then $\hat{F}$ maps a neighbourhood of x in M onto a neighbourhood of $\hat{q}$. This neighbourhood contains points outside the Jordan curve C, i.e. with winding number 0, which contradicts Proposition 3.4. So $\hat{F}^{-1}(\hat{q}) \cap \overline{M} \subset \partial M$, and it is a single point because $\omega \mapsto c(\omega)$ is injective.

All charges. Suppose V_q had five distinct critical points. At most one lies in $\overline{M}$, so at least four lie in $\Delta^\circ \setminus \overline{M}$, where they are nondegenerate saddles. By the implicit function theorem, they persist as

four distinct nondegenerate saddles of $V_{q'}$ for all q' near q. Since C is a curve, we may choose $\hat{q}' \notin C$, and then $V_{q'}$ has at most 3 saddles by Lemma 2.4. This contradiction shows that there are at most four equilibria for every q. In particular there are finitely many, and all are isolated. ∎

6 A local chart in which everything is explicit

6.1 The chart

A positively oriented triangle together with an interior point, up to orientation-preserving similarity, is determined by the angles at the point and by its barycentric coordinates. Precisely, let $\theta_1, \theta_2, \theta_3 \in (0, \pi)$ with $\sum_i \theta_i = 2\pi$, and let λ lie in the open simplex. Put $x_0 = 0$ and

$$\begin{aligned} v_1 = 1, \quad v_2 = e^{i\theta_3}, \quad v_3 = e^{-i\theta_2}, \\ r_i = \frac{\sigma_i}{\lambda_i}, \qquad p_i = r_i v_i. \end{aligned} \tag{6.1}$$

Here $v_i = -u_i$ is the direction from x_0 to p_i, and the angle between v_{i+1} and v_{i+2} is θ_i. By Lami's theorem, the weights w with $\sum_i w_i r_i v_i = 0$ satisfy $w_i r_i \propto \sigma_i$. Hence $\lambda(x_0) = \lambda$, and the Lami constant of Lemma 2.2 is normalized to $c = 1$. Conversely, every pair (Δ, x_0) arises in this way. A cyclic relabelling lets us assume that $\theta_3 = \max_i \theta_i$. We use the rectangular coordinates

$$\theta_3 \in \left[\tfrac{2\pi}{3}, \pi\right], \qquad \tau \in [-1, 1], \qquad \theta_{1,2} = \frac{2\pi - \theta_3}{2} \pm \tau\, \frac{3\theta_3 - 2\pi}{2}, \tag{6.2}$$

which parametrize $\{\theta_3 = \max\}$ exactly. The certification domain is the compact set

$$\mathcal{D} = \left[\tfrac{2\pi}{3}, \pi\right] \times [-1, 1] \times \{(\lambda_1, \lambda_2) : \lambda_1, \lambda_2 \geq 0,\ \lambda_1 + \lambda_2 \leq 1\}, \qquad \lambda_3 = 1 - \lambda_1 - \lambda_2. \tag{6.3}$$

It contains degenerate limits, namely flat configurations ($\theta_3 = \pi$) and $\lambda_i = 0$, which correspond to no actual triangle. Verifying strict inequalities on this larger compact set is stronger than needed. The rescaled formulas are used on the region relevant to $|Z| \leq \kappa$. Singular corner regimes with $|Z| \to 1$ are discarded first, as explained in Section 6.4.

All the conditions are invariant under cyclic relabelling and rotation, except (W), which refers to a labelled vertex. (J) and ν are defined from Z, which is symmetric. The determinant in (I) is invariant under cyclic permutations of coordinates. This is why (W) enters Theorem 5.1 only through the fixed-label statement for Δ_0. We note that there is no positive uniform margin below $\frac{1}{2}$ for all three vertices on the closed flat-limit domain: for the vertex p_3, which is not on the side subtending the largest angle, $|d \arg u_3 / d\omega|$ tends to $\frac{1}{2}$ in the flat limit.

6.2 Derivatives

Let $\rho_i = 1/r_i = \lambda_i/\sigma_i$ and $w_i = p_i - x$. A direct computation from the barycentric formula, using $2|\Delta| = r_1 r_2 r_3$ in the chart, gives the constant Wirtinger derivatives of the barycentric coordinates:

$$\mu_i := \frac{\partial \lambda_i}{\partial z} = \frac{i}{2}\, \rho_i \big(\rho_{i+2}\, \bar{v}_{i+1} - \rho_{i+1}\, \bar{v}_{i+2}\big), \qquad \frac{\partial \lambda_i}{\partial \bar{z}} = \bar{\mu}_i. \tag{6.4}$$

Since $u_i^2 = v_i^2 = w_i/\bar{w}_i$, with $\partial_z (w/\bar{w}) = -1/\bar{w}$ and $\partial_{\bar{z}} (w/\bar{w}) = w/\bar{w}^2$, and since $1/\bar{w}_i = \rho_i v_i$ and $w_i/\bar{w}_i^2 = \rho_i v_i^3$ at x_0, we get

$$\begin{aligned} Z &= \sum_i \lambda_i v_i^2, \\ Z_z &= \sum_i (\mu_i v_i^2 - \lambda_i \rho_i v_i), \qquad Z_{\bar{z}} = \sum_i (\bar{\mu}_i v_i^2 + \lambda_i \rho_i v_i^3). \end{aligned} \tag{6.5}$$

$$
\begin{aligned}
Z_{zz} &= -2\sum_i \mu_i\rho_i v_i,\\
Z_{z\bar z} &= \sum_i (\mu_i\rho_i v_i^3 - \bar\mu_i\rho_i v_i - \lambda_i\rho_i^2 v_i^2),\\
Z_{\bar z\bar z} &= 2\sum_i (\bar\mu_i\rho_i v_i^3 + \lambda_i\rho_i^2 v_i^4).
\end{aligned}
\tag{6.6}
$$

Differentiating $Z_z t + Z_{\bar z}\bar t = iZ$ once more along ∂M gives $Z_z t' + Z_{\bar z}\bar t' = R$, with $R = -Z - \big(Z_{zz}t^2 + 2Z_{z\bar z}|t|^2 + Z_{\bar z\bar z}\bar t^2\big)$. Hence

$$
t' = \frac{\overline{Z_z}\,R - Z_{\bar z}\,\bar R}{J}, \qquad \nu = \frac{d\arg t}{d\omega} = \operatorname{Im}\frac{t'}{t}. \tag{6.7}
$$

The cusp condition $t \parallel k$ is equivalent to $\bar t^{\,2} Z \in (-\infty, 0)$, because $k^2 = -Z/|Z|$. For (W), $d\arg u_1/d\omega = \operatorname{Im}(t\,\bar u_1)/r_1 = -\rho_1 \operatorname{Im} t$, since $u_1 = -v_1 = -1$.

6.3 The inflection function

Fix ω_0 and rescale charge space by $D = \operatorname{diag}(q(\omega_0))$, with $q = F(x_0)$. In the chart, $\lambda_i r_i = \sigma_i$. Differentiating (3.6), using $du_i/d\omega = (t - (u_i\cdot t)u_i)/r_i$ and $dr_i/d\omega = u_i\cdot t$, gives $Dm = -s\,a$ and $Dm' = -s\,b$ at ω_0, where

$$
a_i = \sigma_i\,(u_i\cdot k), \qquad b_i = \sigma_i\,(u_i\cdot k') - \lambda_i\big[(s+2)(u_i\cdot t)(u_i\cdot k) - t\cdot k\big], \qquad k' = \tfrac{i}{2}k. \tag{6.8}
$$

Since $(Du)\times(Dv) = \det D\cdot D^{-1}(u\times v)$, condition (I) at ω_0 is equivalent to $a\times b \neq 0$. Two identities simplify this. First, $\sum_i a_i = 0$ by Lami, since $\sum_i \sigma_i u_i = \sum_i \lambda_i(x_0 - p_i) = 0$. Second, $\sum_i b_i = 0$: the first terms sum to zero for the same reason, and $\sum_i \lambda_i (u_i\cdot t)(u_i\cdot k) = t^{\mathsf T} T k = \frac{1}{s+2}\,t\cdot k$, because k is the eigenvector of T with eigenvalue $\frac12(1-\kappa) = \frac{1}{s+2}$. For vectors with zero sum, $\det(\mathbf 1, a, b) = 3(a_3b_1 - a_1b_3)$, and $a\times b \parallel \mathbf 1$. So (I) is equivalent to $a_3b_1 - a_1b_3 \neq 0$. Expanding, and using $(u_3\cdot k)(u_1\cdot ik) - (u_1\cdot k)(u_3\cdot ik) = \operatorname{Im}(u_1\bar u_3) = \sin\theta_2$, we obtain

$$
\begin{aligned}
\check I := \frac{a_3b_1 - a_1b_3}{\sigma_3} &= \tfrac12\sigma_1\sigma_2 + (s+2)\,Q(v_1,v_3)\big[\lambda_1\sigma_3(v_1\cdot\check t) - \sigma_1\lambda_3(v_3\cdot\check t)\big]\\
&\quad - \lambda_1\sigma_3\,Q(v_3,\check t) + \sigma_1\lambda_3\,Q(v_1,\check t),
\end{aligned}
\tag{6.9}
$$

where $\check t = t/\sigma_3$ and

$$
Q(a,b) = (a\cdot k)(b\cdot k) = \tfrac12\operatorname{Re}(a\bar b) + \tfrac12\operatorname{Re}\big(ab\,\bar k^2\big), \qquad \bar k^2 = -\bar Z/|Z|. \tag{6.10}
$$

Thus $\det(\mathbf 1, m, m') \neq 0$ if and only if $\check I \neq 0$, and $\det(\mathbf 1, a, b) = 3\sigma_3\check I$. We verify the stronger sign condition $\check I > 0$. The identity $\det(\mathbf 1, a, b) = 3\sigma_3\check I$ was also checked numerically, against a chart-free implementation, to relative accuracy 10^{-15}.

6.4 Rescaling for the flat face

As $\theta_3 \to \pi$, $\rho_3 = \lambda_3/\sigma_3$ blows up. We therefore multiply all inverse lengths by σ_3: $\hat\rho_i = \sigma_3\rho_i$, so that $\hat\rho_3 = \lambda_3$, and we write $\sigma_3\rho_3\rho_j = \lambda_3\rho_j$. Then $\sigma_3\mu_i$, $\sigma_3 Z_z$, $\sigma_3 Z_{\bar z}$ and $\sigma_3^2 Z_{zz}, \dots$ are polynomials in $\sin\theta_i$, $\cos\theta_i$, λ_i, ρ_1 and ρ_2, in which ρ_3 no longer appears. The vector $\check t = t/\sigma_3$, the rate ν and $\check I$ are all computed from these rescaled quantities. The quantities ρ_1, ρ_2 can still blow up as σ_1 or $\sigma_2 \to 0$. This happens only when $\theta_3 \to \pi$ and $\tau \to \pm 1$, where $|Z| \to 1$, so such points are discarded by the test $|Z| > \kappa$ before these quantities are needed.

7 Interval verification

7.1 Arithmetic

We use intervals of IEEE 754 binary64 numbers [8] and complex rectangles. After each operation $+, -, \times, \div, \sqrt{}$ under round-to-nearest, the result is moved outward by one unit in the last place. Since the correctly rounded result is within half an ulp of the exact one, this encloses the exact result; overflow and underflow are also handled correctly. The constant π is enclosed by two consecutive doubles.

To enclose sin and cos at a double x, we reduce by an interval multiple of $\pi/2$ to $|y| \le 0.8$. We then evaluate the Taylor polynomials of degree 21 and 20 in interval arithmetic, with interval-enclosed coefficients, and add the Lagrange remainders $|y|^{23}/23!$ and $|y|^{22}/22!$, bounded above by $|y|^{23}/(2.5 \cdot 10^{22})$ and $|y|^{22}/(1.1 \cdot 10^{21})$. Interval extensions on $[x_-, x_+]$ take the hull of the endpoint enclosures. Any extremum ± 1 whose abscissa may lie in the interval is added; the abscissa is tested with a margin of 10^{-9}, which can only widen the result. No accuracy of the platform's `libm` is assumed. All powers in these remainder bounds are themselves evaluated by outward-rounded interval products. Range reduction multiplies an interval for π by the signed integer, so negative multiples preserve endpoint order. The point enclosures were compared with 80-digit `mpmath` values, including points adjacent to quadrant boundaries, with no violation.

For a box B with midpoint c, we use the mean-value form

$$f(B) \subseteq f(c) + \sum_{j=1}^{4} \partial_j f(B)\,(B_j - c_j), \tag{7.1}$$

intersected with the naive interval evaluation $f(B)$. Here $\partial_j f(B)$ is obtained by forward-mode automatic differentiation in interval arithmetic, and $B_j - c_j$ is itself enclosed rigorously.

7.2 Branch and bound

The domain $\mathcal{D}$ is covered by an initial grid. For the simplex, we use the exact rational cells $[i/n, (i+1)/n] \times [j/n, (j+1)/n]$ with $0 \le i, j < n$ and $i + j < n$. Their union covers the closed simplex, including its diagonal by closure. Each rational endpoint is enclosed by outward rounding before the floating-point boxes are formed; this prevents the thin gaps that an uncorrected `linspace` grid can leave at $\lambda_1 + \lambda_2 = 1$. Exact rational regression tests check this inclusion. The lower end of the θ_3 range is moved outward below $2\pi/3$, and the upper end encloses π. For each box B we compute enclosures of $|Z|^2$, J, $\check{I}$, ν, $\vec{t}^{\,2} Z$ and $\rho_1 \operatorname{Im} t$, all in the rescaled form. A box is *closed* if one of the following holds; each is a strict inequality between enclosure endpoints, so a NaN can never close a box:

(i) $|Z|^2 > \kappa^2 (1 + 10^{-12})$ on B: no point of $\overline{M}$;

(ii) $|Z|^2 < \kappa^2 (1 - 10^{-12})$ and $J < 0$ on B: interior points, where (J) holds;

(iii) $J < 0$, $\check{I} > 0$, $|\rho_1 \operatorname{Im} t| < \tfrac{1}{2}$ on B, and either $\nu < \tfrac{1}{2}$ on B or no cusp is possible in B, i.e. $\operatorname{Im}(\vec{t}^{\,2} Z)$ has constant strict sign or $\operatorname{Re}(\vec{t}^{\,2} Z) > 0$ on B.

Boxes not closed are bisected. The bisection direction is chosen heuristically, as the coordinate contributing most to the width of the first failing enclosure; this choice affects only efficiency. Boxes narrower than 10^{-9} in every coordinate, or still open after the iteration limit, would have been reported as unresolved. A cheap naive enclosure of $|Z|^2$ prefilters test (i).

7.3 Results

	run A	run B
initial grid ($\theta_3 \times \tau \times \lambda$)	$12 \times 12 \times 78$	$10 \times 10 \times 55$
initial boxes	11 232	5 500
boxes processed	9 965 924	9 966 320
closed by (i) (outside $\overline{M}$)	1 023 498	1 030 113
closed by (ii) (interior)	837 622	839 270
closed by (iii) (boundary)	3 127 458	3 116 527
of which with no cusp possible	2 430 074	2 434 734
unresolved boxes	**0**	**0**
largest upper bound of $\sigma_3^2 J$ on closed boxes	$-2.08 \cdot 10^{-4}$	$-1.56 \cdot 10^{-4}$
smallest lower bound of $\check{I}$	$3.21 \cdot 10^{-7}$	$1.52 \cdot 10^{-7}$
largest upper bound of ν where a cusp is possible	0.4999988	0.4999994
largest bound of $\lvert d \arg u_1 / d\omega \rvert$	0.377190	0.387974
wall-clock time (16 processes)	474.5 s	496.8 s

Table 1. Certification of Theorem 5.1, items 1–2, for $s = 1$. The two runs start from different subdivisions of $\mathcal{D}$. In each run the counts are consistent with a binary tree: processed $= 2 \times$ (closed) $-$ initial. The bounds in the lower rows are those of the enclosures at the moment each box was closed, rounded conservatively for display; they are not the true margins. Full precision values, source hashes and logs are retained with the certificate files.

A box is closed as soon as its enclosure clears the threshold, so the extreme bounds in Table 1 come from boxes that were barely closed. They do not measure how close the true quantities come to the thresholds. Sampling several hundred thousand points of ∂M directly in the chart, including near the flat face $\theta_3 = \pi$ and the faces $\lambda_i = 0$, gives $\check{I} \geq 0.097$, $\sigma_3^2 J \leq -0.098$, $\nu \approx -1$ at numerically located cusp points, and $\lvert d \arg u_1 / d\omega \rvert \leq 0.29$. All of these are well inside the thresholds. These samples are diagnostics, not certified margins. The proof uses the interval decisions for every box and the verified coverage of the exact domain; termination is recorded by a zero unresolved count.

The base triangle. Item 3 of Theorem 5.1 is checked in barycentric coordinates (λ_1, λ_2) on Δ_0. Boxes with some $\lambda_i > \frac{2}{3}$ are discarded analytically, since there $\lvert Z \rvert \geq 2\lambda_i - 1 > \frac{1}{3}$. Every other box must satisfy either $\lvert Z \rvert^2 > \kappa^2$, or $\theta_3 > \theta_1$ and $\theta_3 > \theta_2$. The latter are evaluated as $(w_1 \cdot w_2)\, r_3 < (w_2 \cdot w_3)\, r_1$ and $(w_1 \cdot w_2)\, r_3 < (w_1 \cdot w_3)\, r_2$, with $w_i = p_i - x$. Starting from 2 080 boxes, the bisection processed 2 356 boxes with none unresolved. A second run, starting from 1 176 boxes, processed 1 618 boxes with none unresolved. Numerically, the smallest value of $\theta_3 - \max(\theta_1, \theta_2)$ on $\partial M(\Delta_0)$ is 0.85.

7.4 Validation of the implementation

Besides the internal consistency of the two runs, the closed forms were checked in four ways.

(i) Against an independent implementation that obtains the derivatives of λ by solving linear systems, to 10^{-14}.

(ii) Against chart-free formulas computed directly from the coordinates of a triangle, to 10^{-15}.

(iii) Against finite differences along numerically traced curves ∂M of concrete triangles.

(iv) Enclosure tests: at about $3 \cdot 10^5$ random points of random boxes, for three exponents, the interval enclosures of $|Z|^2$, J, $\check{I}$ and ν contain the floating-point values of the independent implementation.

8 Further evidence and extensions

8.1 A theory-free check

Independently of the reduction, we counted planar equilibria by multistart damped Newton iteration on ∇V_q. The starts were a 70×70 barycentric grid plus multiscale clusters near each vertex. We took 64 000 random configurations. The triangles $(0, 1, a + ib)$ had $0 \le a \le \frac{1}{2}$, b log-uniform in $[10^{-2.5}, 1]$, and $p_1 p_2$ the longest side. The charges were log-uniform over six decades. Newton found 2 equilibria in 63 975 cases and 4 in 20 cases, and never 6. Five extreme cases gave odd counts (1 or 5), which Lemma 2.4 forbids. Re-examining them by numerical evaluation of the winding-number criterion of Proposition 3.4 was consistent with two equilibria in all five cases; Newton had missed or duplicated points. The rarity of the value 4 reflects the small charge-space area enclosed by C under log-uniform sampling (Figure 2).

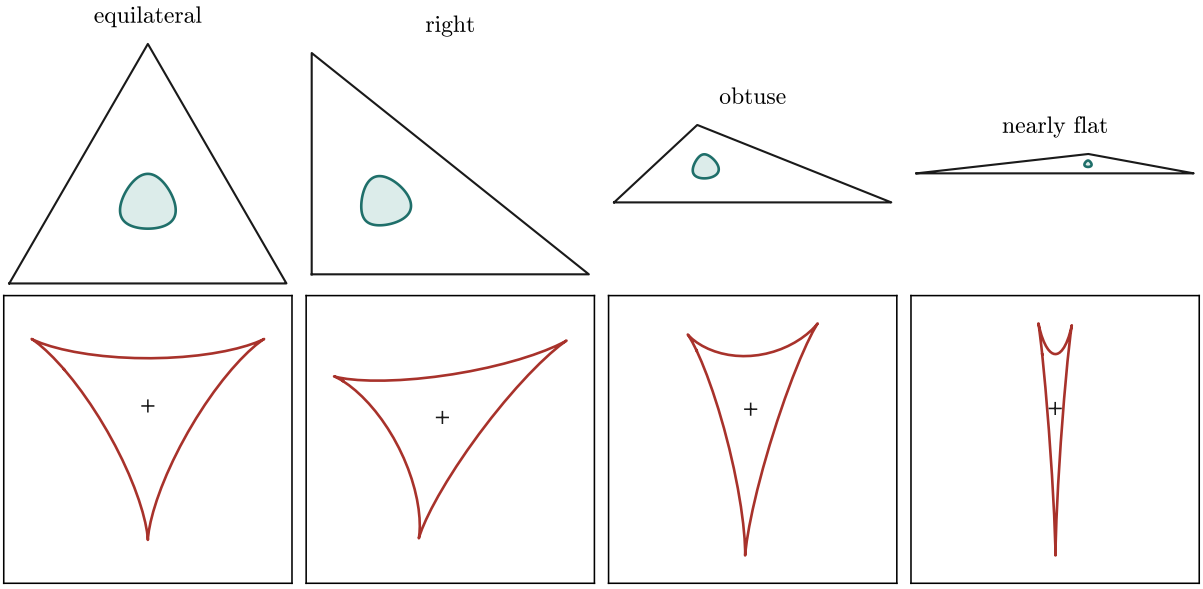


Figure 2. Top: the minimum region M (shaded) for four triangle shapes. Bottom: the corresponding caustics C, each in the projective chart centred at the image $(+)$ of the point where $Z = 0$. All are simple curves with three cusps and no inflections, as predicted. The four-equilibria region is the interior of C. As the triangle flattens, C becomes a thin needle.

8.2 Other exponents

The arguments of Sections 2–5 hold for every $s > 0$, with $\kappa = s/(s + 2)$, once the conditions of Theorem 5.1 are verified for that s. Numerically, the structure required by Theorem 4.3 persists for all triangle shapes we sampled when $s \lesssim 1.5$, so the same certification should succeed there. Above a shape-dependent threshold, about 2.13 for the equilateral triangle and about 1.53 for nearly degenerate shapes, the caustic acquires pairs of inflections, and the argument above no longer applies as stated. We have nevertheless never observed a winding number 2, i.e. a configuration with 6 equilibria, for any s we tried.

8.3 Charges of mixed signs

Let $q = (+,+,-)$; the other mixed pattern reduces to this by a global sign change. Lemma 2.1(a) still confines equilibria to Π. By the argument of Lemma 2.2, an equilibrium has barycentric coordinates of the sign pattern of $\pm q$. It therefore lies in $R_A = \{\lambda_1, \lambda_2 > 0 > \lambda_3\}$ or in $R_B = \{\lambda_1, \lambda_2 < 0 < \lambda_3\}$, and the Hessian is $\pm sW[(s+2)T - I]$, with T as before.

Proposition 8.1. *For $q_1, q_2 > 0 > q_3$, $V|_\Pi$ has no local maxima, and all its nondegenerate local minima lie in $M_A = R_A \cap \{|Z| < \kappa\}$.*

Proof. In R_A the Hessian is a positive multiple of $(s+2)T - I$. Since $\operatorname{tr} T = 1$ and $2/(s+2) < 1$, the larger eigenvalue is positive, so there is no maximum. In R_B it is a positive multiple of $I - (s+2)T$. There p_3 lies inside the triangle (x, p_1, p_2), and for $e \perp u_3$ we get $e^{\mathsf{T}} T e = \lambda_1 (u_1 \cdot e)^2 + \lambda_2 (u_2 \cdot e)^2 \le 0$. So $I - (s+2)T$ has a positive eigenvalue, while its trace is $-s < 0$, so it also has a negative eigenvalue. Hence R_B contains only saddles, and nondegenerate minima in R_A are characterized by $|Z| < \kappa$ exactly as before. ∎

For Morse configurations with $\sum_i q_i \neq 0$, Poincaré–Hopf again gives $2 + 2m$ equilibria, so Maxwell's bound for mixed signs is again equivalent to $m \le 1$. We measured the analogues of (J), (I), (T) along ∂M_A for 38 random triangles, including very flat and nearly collinear ones, using chart-free formulas. In all cases $J < 0$, there were no inflections, the turning rate of ∂M_A was nonpositive, and the caustic had exactly three cusps. So Theorem 4.3 would again give $m \le 1$. A certificate requires a compactification of the chart, which is non-compact here because λ is unbounded on R_A. It must handle three degenerate regimes: flat $(+,-,+)$, flat $(+,+,-)$, and the far-field regime $\sum_i q_i \approx 0$. It must also handle the fact that the shortcut (W) fails in the far field. We have not done this.

9 Discussion

What has been proved. Modulo the correctness of the computation behind Theorem 5.1, three positive Coulomb charges have at most four equilibria, generically 2 or 4, with the dichotomy decided by the position of the normalized charge vector relative to the caustic C. The argument is geometric. The equilibrium count is the winding number of a caustic, and that winding number is bounded by a Sturm-type comparison for fronts. The computer is used only to verify four open conditions on a compact set of shapes and one condition on a single triangle.

What an independent check should cover.

1. The arguments of Sections 2–5, especially Proposition 3.2, Lemmas 4.1–4.2, and the passage from (T) and (I) to the hypotheses of Theorem 4.3 in Proposition 5.2.
2. The closed forms of Section 6 and their rescaled versions. These are short and can be re-derived symbolically.
3. The interval library and the automatic differentiation. These are small (about 400 lines) but home-made. A re-implementation with a standard rigorous library such as Arb or MPFI, or in a proof assistant, would be valuable.
4. The coverage of the domain: the initial grid, the symmetry reduction, and the rounding at $\theta_3 = 2\pi/3$ and $\theta_3 = \pi$.

Outlook. The method should apply verbatim to Riesz exponents $s \lesssim 1.5$, where the required structure is observed numerically, and it should extend to charges of mixed signs after a compactification of the chart. For larger numbers of charges, the present planar-front argument has no direct analogue; the dimensions and codimensions of the distance surface and its normal spaces must be treated separately. The counterexample of [1] shows that some such step must fail for $n = 5$.

AI-assistance statement

This work was carried out with extensive assistance from Claude (Anthropic, model Claude Opus 5.5), an AI system. The AI system proposed the approach, derived the reduction, the Sturm-type lemma and the closed forms, wrote the verification code, ran the computations and drafted this manuscript, working interactively with the author. A subsequent review by a second AI system (GPT-6-Astra, run in OpenAI Codex) found and corrected three defects in the interval verification: incomplete coverage of the domain by rounding-error-wide slivers, an inverted interval in one angle reduction, and a Taylor remainder computed with the platform power function. It also reran the certificates and revised the exposition and references; Claude then reproduced the main certificate run with identical box counts. This is not a substitute for review by an independent human expert; such a review has not yet been completed. We therefore present this work as a preprint inviting scrutiny, and the accompanying source is provided in full for that purpose.

Code availability

All code, certificate summaries and run logs are in the project repository [16]. The recorded runs use CPython 3.11 and NumPy 2.4.6; the high-precision tests use mpmath 1.3.0. Each certificate includes source SHA256 hashes, parameters and platform versions. A nonzero unresolved count causes a failing exit status. The runner retains logs in a fresh directory and checks hashes and results:

```
python -m pip install -r requirements-review.txt
python reproduce.py --workers 16
```

The two main subdivisions share the same interval implementation. Their agreement is a reproducibility check, not an independent arithmetic proof. The JSON files summarize the executions; they are not leaf-by-leaf certificates for a separate checker. A second rigorous implementation remains desirable.

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